

RUB

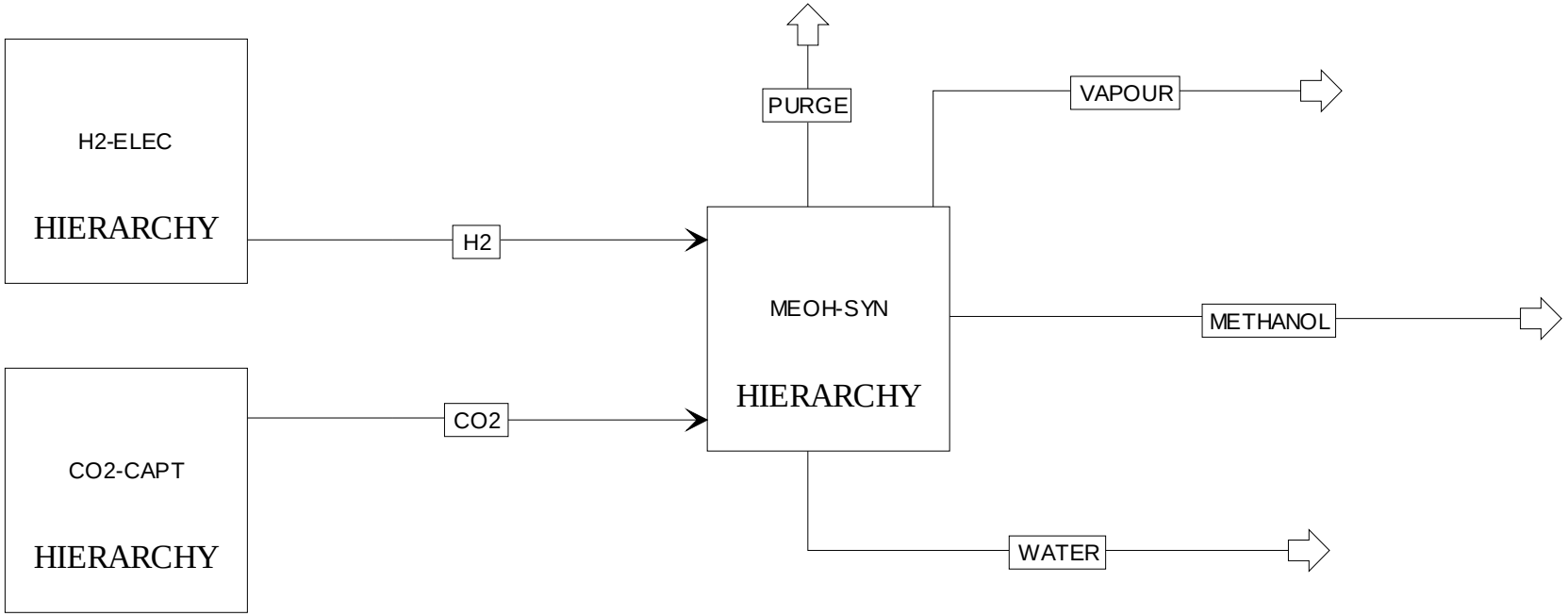
RUHR-UNIVERSITÄT BOCHUM

Green Methanol Production via Power-to-X

Fakultät Maschinenbau
fortschritt studieren

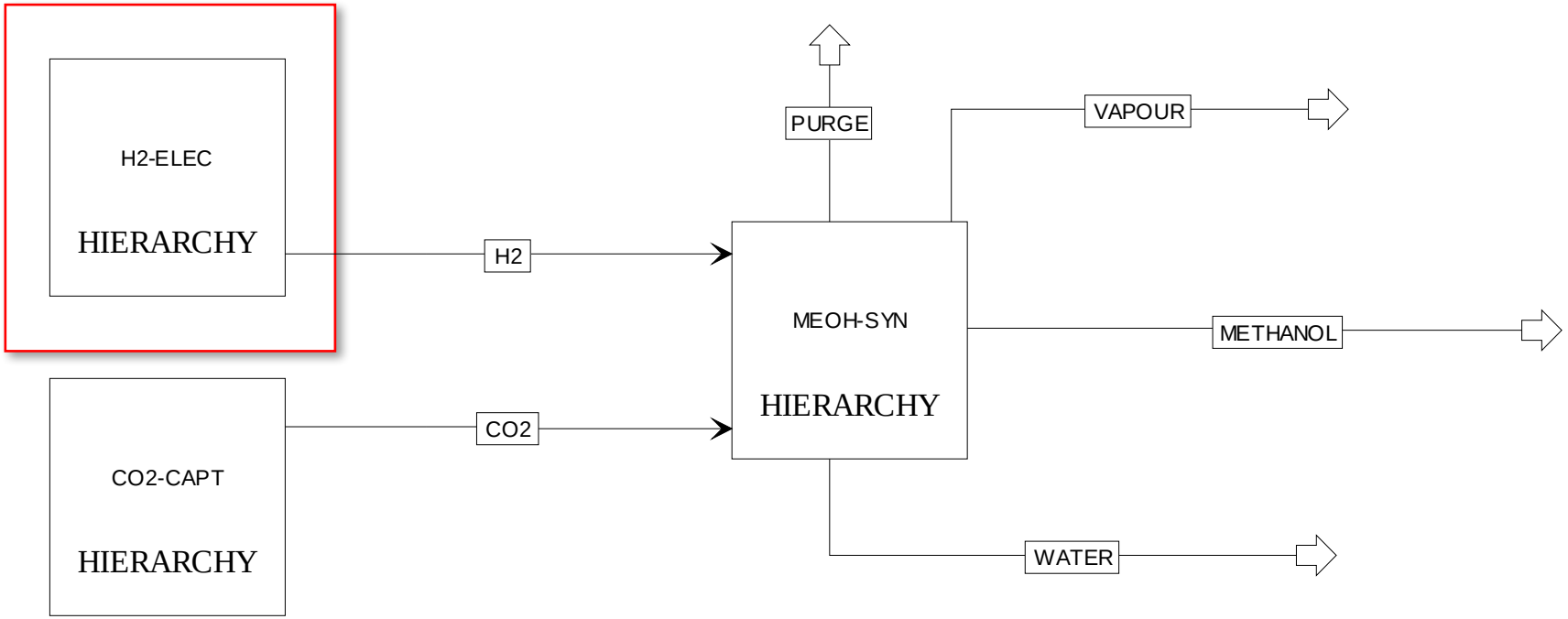
Jonas Bankwitz, Naveed Ahmed, Hamide Aveder | Computer-Aided Process Design | 09.10.2026

Power-to-Methanol — Process Overview

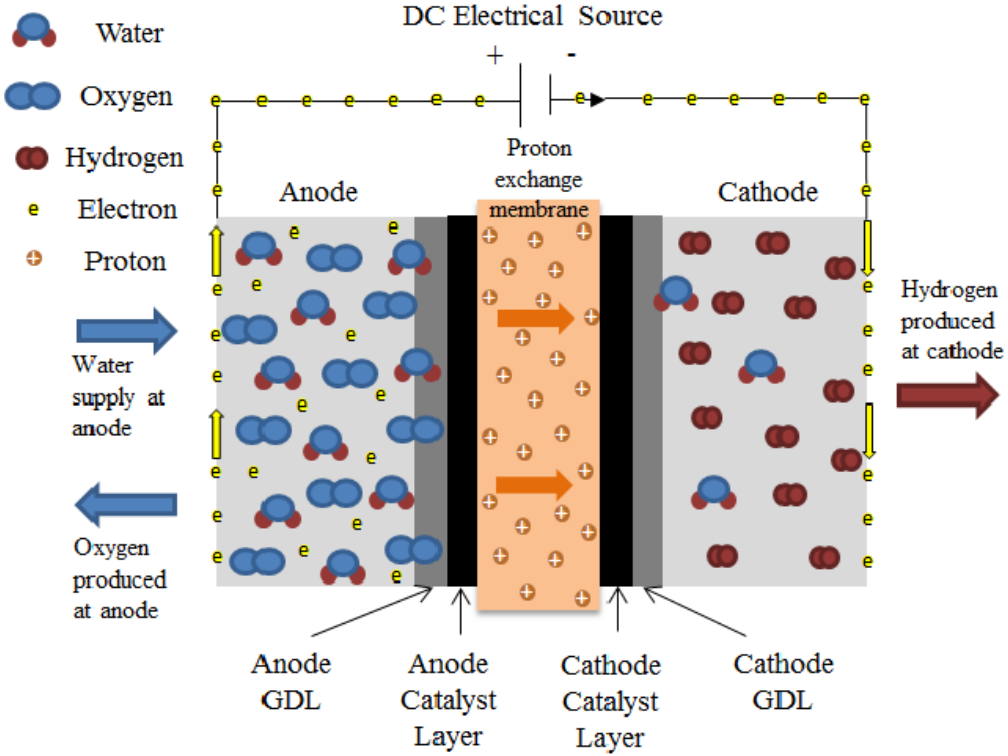


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Hydrogen Electrolysis (PEM)



PEM Electrolysis — Working Principle



Net Reaction:

$$H_2O \rightarrow H_2 + \frac{1}{2}O_2$$

Anode:

$$H_2O \rightarrow \frac{1}{2}O_2 + 2H^+ + 2e^-$$

Cathode:

$$2H^+ + 2e^- \rightarrow H_2$$

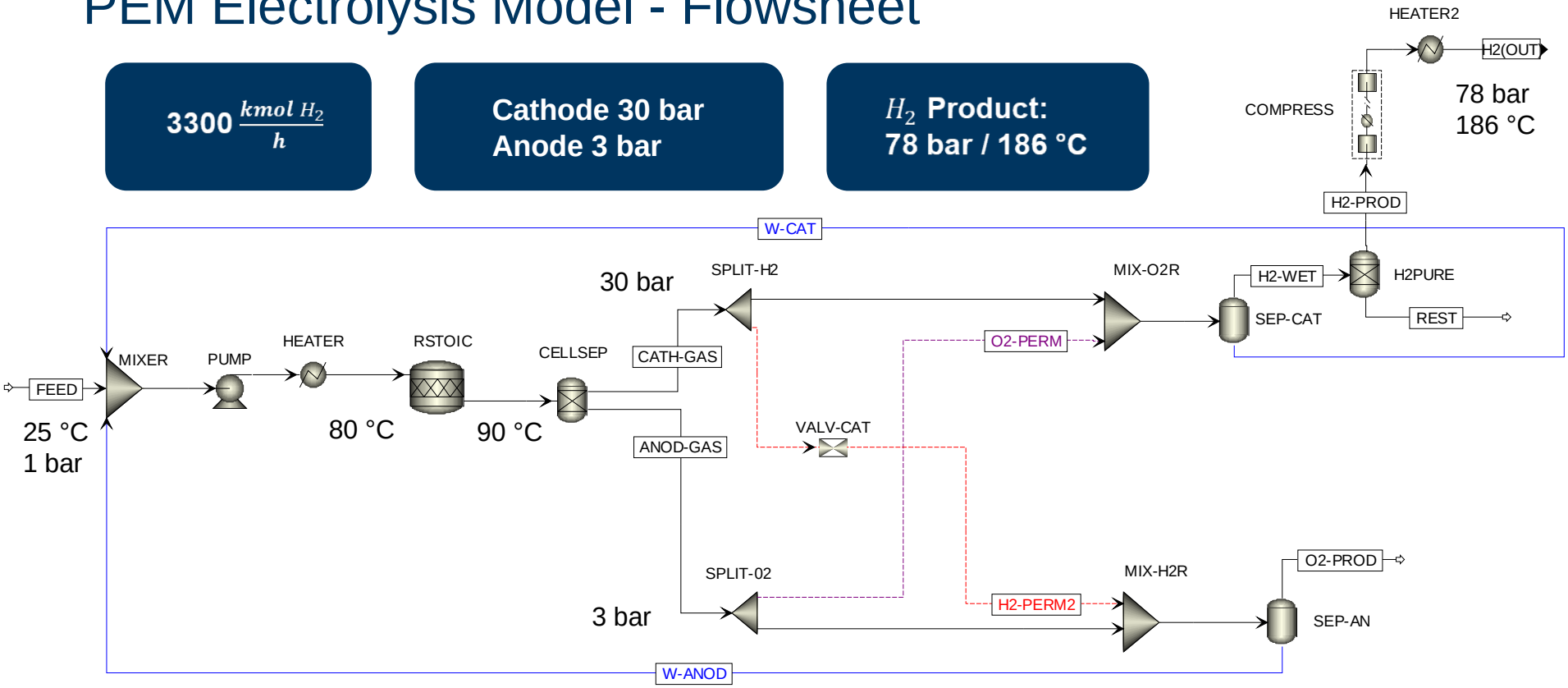
https://www.researchgate.net/figure/A-PEM-electrolysis-cell_fig2_374390953

PEM Electrolysis Model - Flowsheet

3300 $\frac{kmol\ H_2}{h}$

Cathode 30 bar
Anode 3 bar

H_2 Product:
78 bar / 186 °C



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Design Choices

Property Method

- NRTL + Henry
(water + dissolved gases)

Stack

- Rstoic reactor
(mass and energy balance)

Membrane leak included

- Crossover is modelled

Differential Pressure

- 30 bar cathode
- 3 bar anode
- (10 % rule)

Water recycled

- Fresh demand only
- 9.09 kg/kg

Delivery

- Compressed to 78 bar
- heated to 186 °C for synthesis

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Model Verification

Reaction Stoichiometry

Water consumed	3305.28 kmol / h
H_2 formed	3305.28 kmol / h
O_2 formed	1652.6 kmol / h

$$H_2O \rightarrow H_2 + \frac{1}{2}O_2$$ ✔

Mass balance closure

Fresh water in	3356.7 kmol / h
Water out & consumed	3356.8 kmol / h
Error	0.005 %

Flowsheet correctly connected ✔

Design specification

Design Spec target	3300 kmol / h
Simulated H_2 product	3300 kmol / h
Recycle loop	converged

Specification and tears solved ✔

Energy balance consistency

Reaction Duty (Aspen)	262764 kW
Hand check n*ΔH	262011 kW
Deviation	0.29 %

Matches thermodynamic principles ✔

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Model Validation

Quantity	This Work	Zaccara et al. (2020)	Deviation
Required energy	53.2 kWh / kg H_2	54.8 kWh / kg H_2	-2.9%
Water demand	9.09 kg H_2O / kg H_2	9.8 kg H_2O / kg H_2	−7.2%
Cathode composition	95.1 mol-% H_2 4.8 mol-% H_2O 0.1 mol-% O_2	95.2 mol-% H_2 4.7 mol-% H_2O 0.1 mol-% O_2	±0.1pp

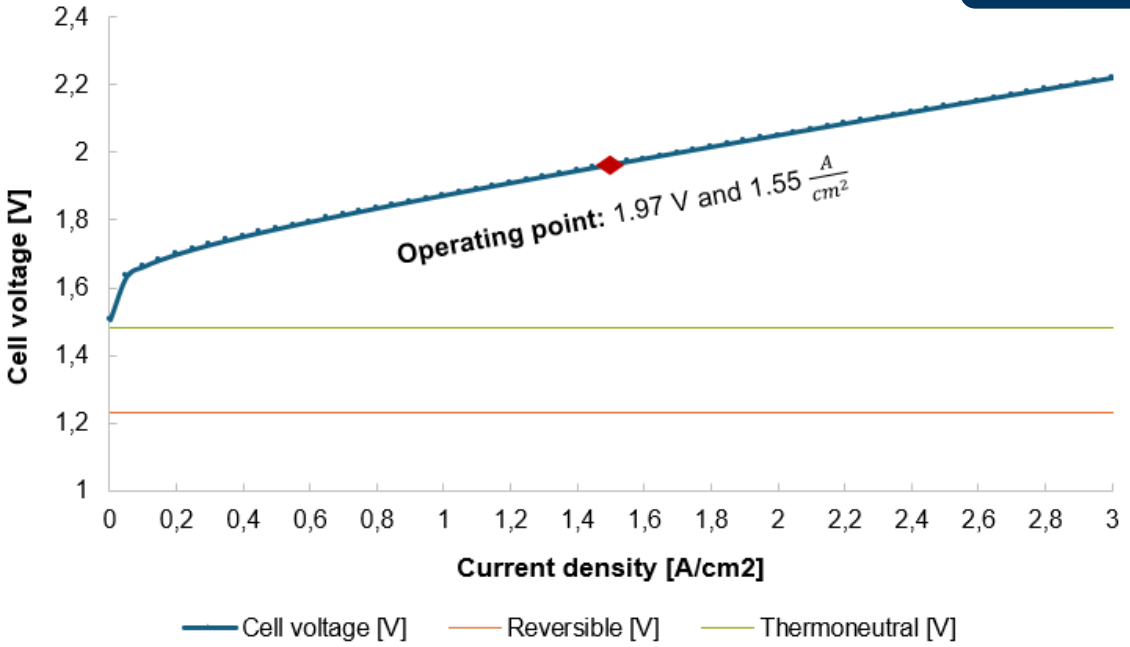
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Model is validated ✔

Cell Polarization Curve

$$U = U_{rev} + \eta_{activation} + \eta_{ohmic} + \eta_{mtransport}$$

Parameters from García-Valverde et al. (2012), evaluated at 80 °C



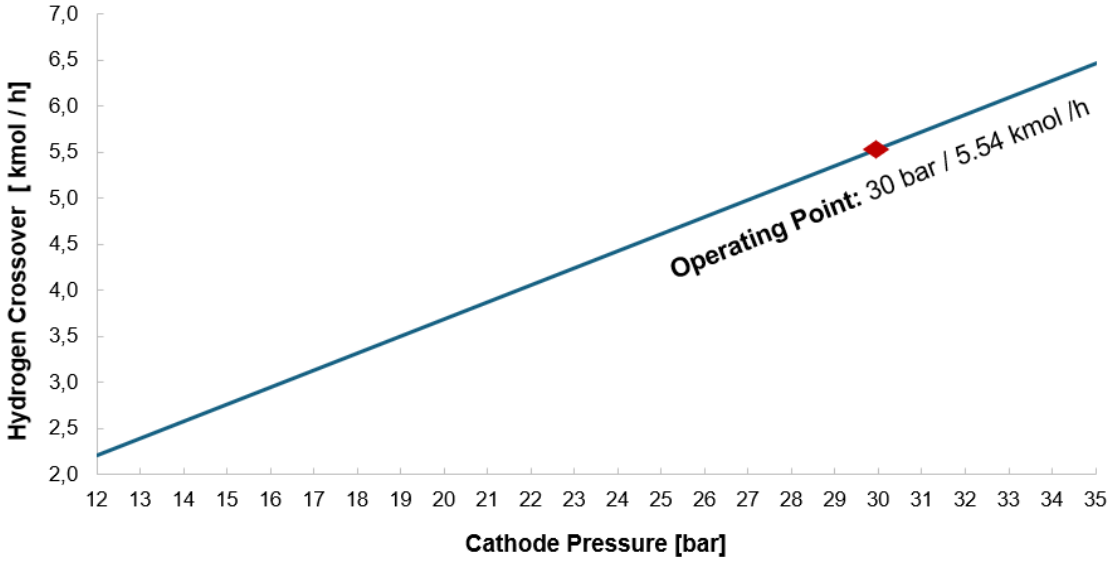
State of the Art Parameters	
Current density (A/cm²)	0.6 - 2.0
Cell Voltage (V)	1.8 - 2.2

(Carmo et al, 2013.)

High current density
➢ small stack area

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Cathode Pressure Trade-off in the PEM Stack



Benefit
- H_2 leaves the stack at 30 bar
- Compressor only lifts from 30 to 78

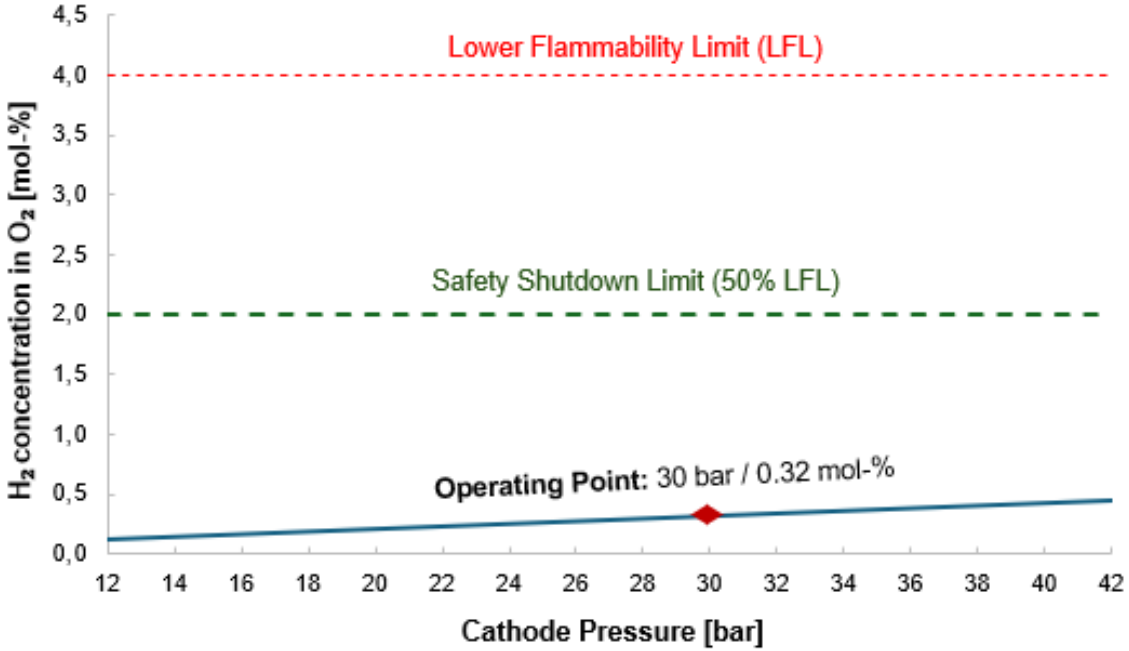
Energy
- Less compression work
- Net lower energy per kg H_2

Limits:
- Crossover rises linearly with pressure
- Membrane strength: PEM range <35 bar

Zaccara et al. (2020)

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Hydrogen Crossover & Explosion Safety



At 30 bar, full load:

- 0.32 mol-% H_2 in O_2
  - 6× below shutdown (2 %)
  - 12× below LFL (4 %)

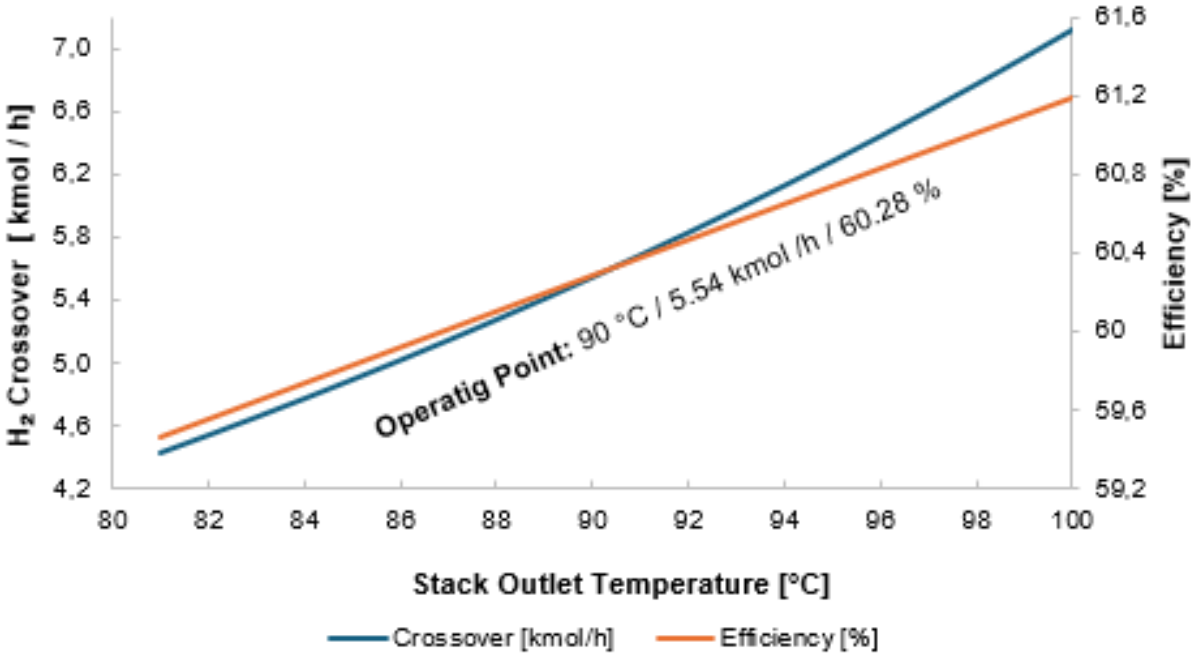
Critical case: part load

- less O_2, same crossover
- sets minimum stack load

Molnarne & Schröder (2019)

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Temperature Effect on Efficiency and Crossover



Efficiency
Higher T → faster kinetics & better proton conductivity → lower cell voltage → higher efficiency

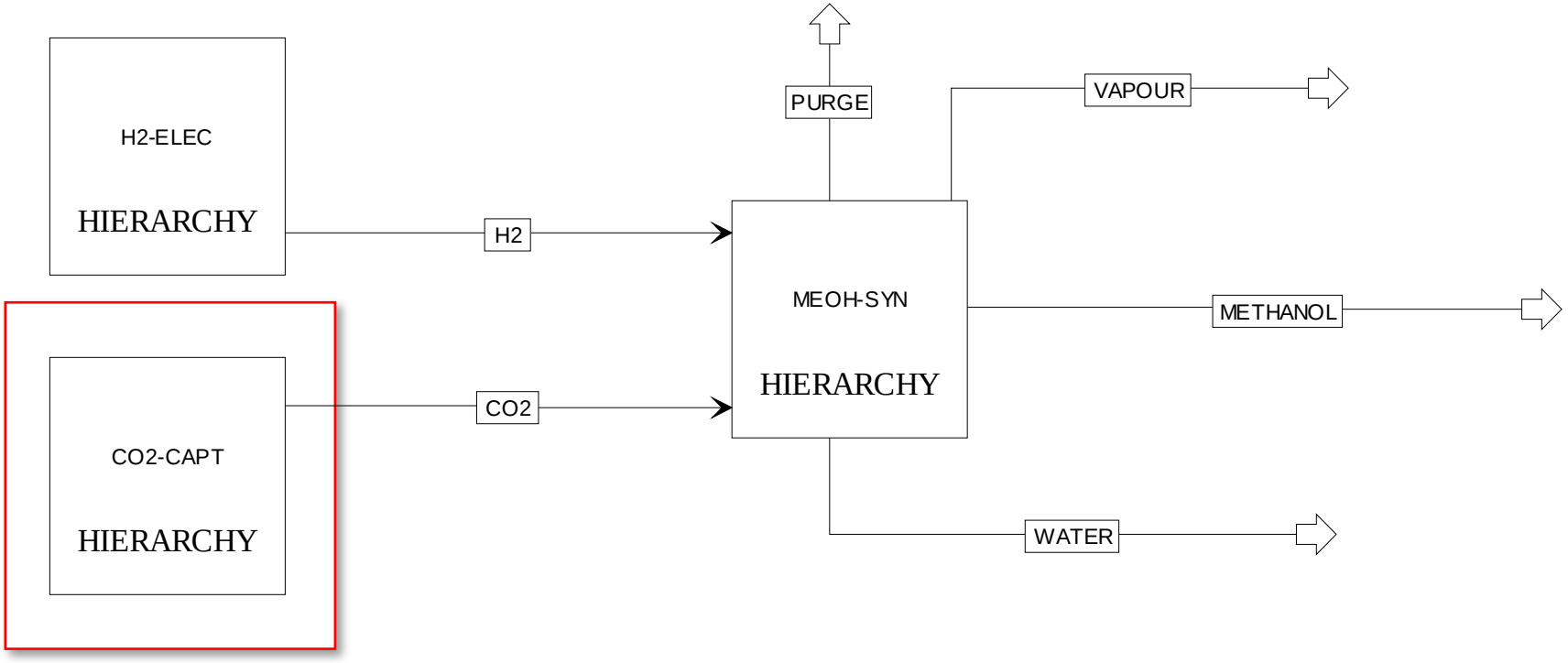
Crossover
Higher T → faster H_2 diffusion → more crossover

Operating Point
Inlet: 80 °C / Outlet: 90 °C
- Low T: proton conductivity drops
- High T: membrane dehydration & degradation

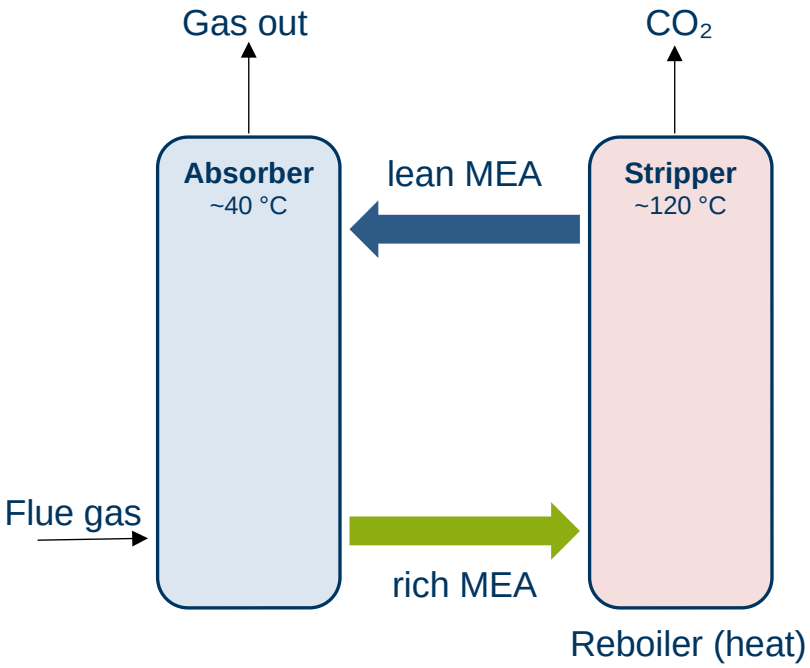
Sandoval-Amador et al. (2026) & Zaccara et al. (2020)

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CO_2 Capture


H2-ELEC
HIERARCHY
CO2-CAPT
HIERARCHY
H2
CO2
PURGE
VAPOUR
MEOH-SYN
HIERARCHY
METHANOL
WATER

CO_2 Capture – Working Principle



Absorption – Absorber, ~40 °C (exothermic)

- CO_2 + 2 MEA → $MEACOO^-$ + $MEAH^+$
  - CO_2 chemically bound as carbamate

Regeneration – Stripper, ~120 °C (endothermic)

- $MEACOO^-$ + $MEAH^+$ → CO_2 + 2 MEA
  - heat reverses the reaction and releases CO_2

Principle – temperature swing, cyclic solvent

- lean MEA binds CO_2 → rich MEA (cold)
- reboiler heat strips CO_2 → lean MEA (hot)
- amine reused; ~2 MEA per CO_2 → loading ≈ 0.5

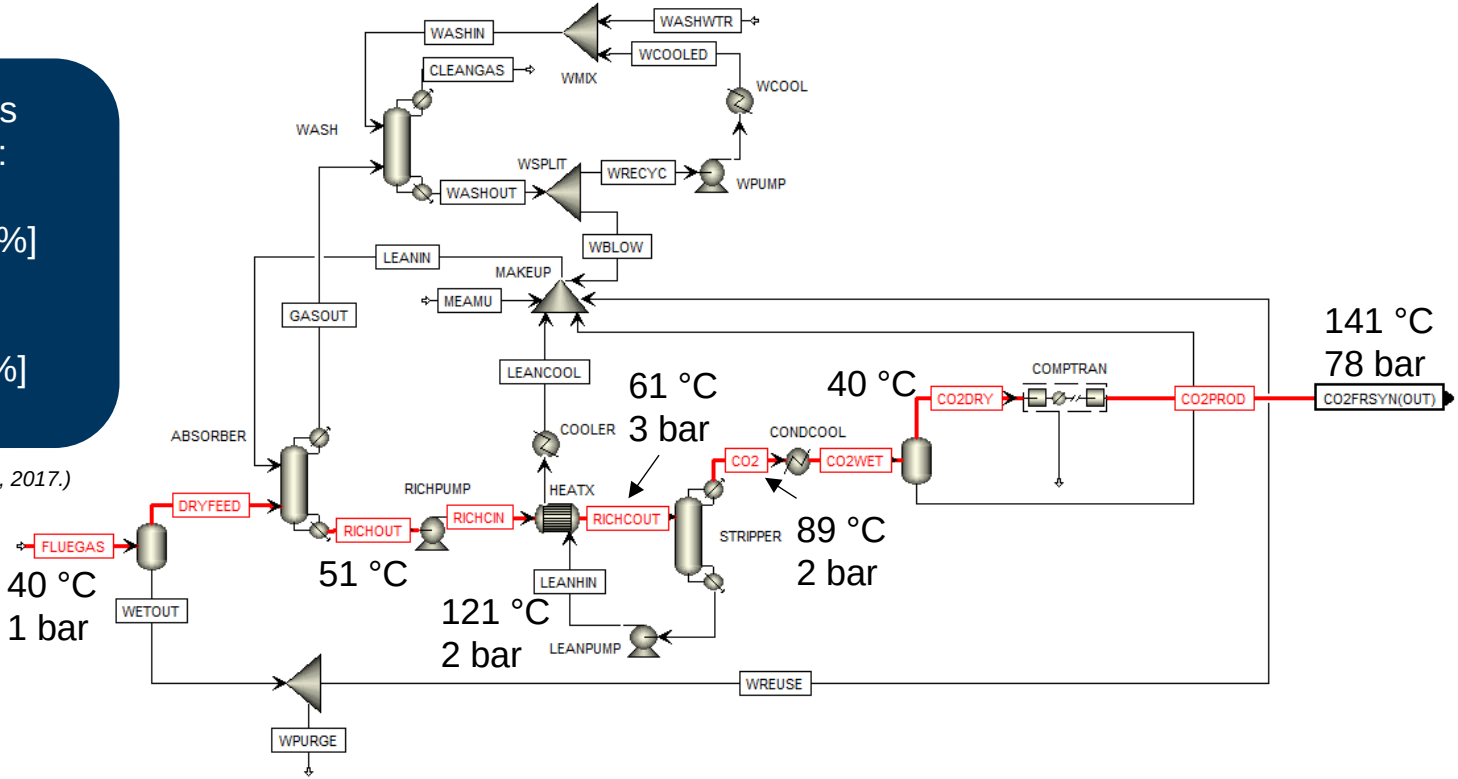
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CO_2 Capture - Flowsheet

CO_2 source: biomass combustion flue gas:

CO2 → 14.35 [mol %]
N2 → 68.29 [mol %]
O2 → 2.38 [mol %]
H2O → 14.17 [mol %]
Ar → 0.81 [mol %]

(Ali et al, 2017.)



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Design Choices

Property Method

- ELECNRTL (Ionic system)
- PENG-ROB (CO2 Compression)

Absorber

- Rate-based
- 20 stages
- Mellapak 250Y packing

Stripper

- Equilibrium
- 12 stages
- Stipper pressure 1.8 bar

Solvent

- Monoethanolamine MEA (~30 wt%)
  - benchmark amine
  - fast absorption

Wash section

- 3 stages
- MEA recovery / emission control

Heat Integration

- Lean / rich exchanger
- Recovers heat from the hot lean solvent → lower reboiler duty

Model Verification

CO_2 mass balance

CO_2 in (fluegas)	1222.2 kmol / h
CO_2 captured	1096.3 kmol / h
CO_2 vented	125.5 kmol / h

Mass balance closes ✔

Overall mass balance

in	1,146,180 kg/h
out	1,146,180 kg/h
Rel. difference	$4*10^{-11}$

In=out ✔

Convergence

Absorber	converged
Stripper	converged
Wash / recycle	converged

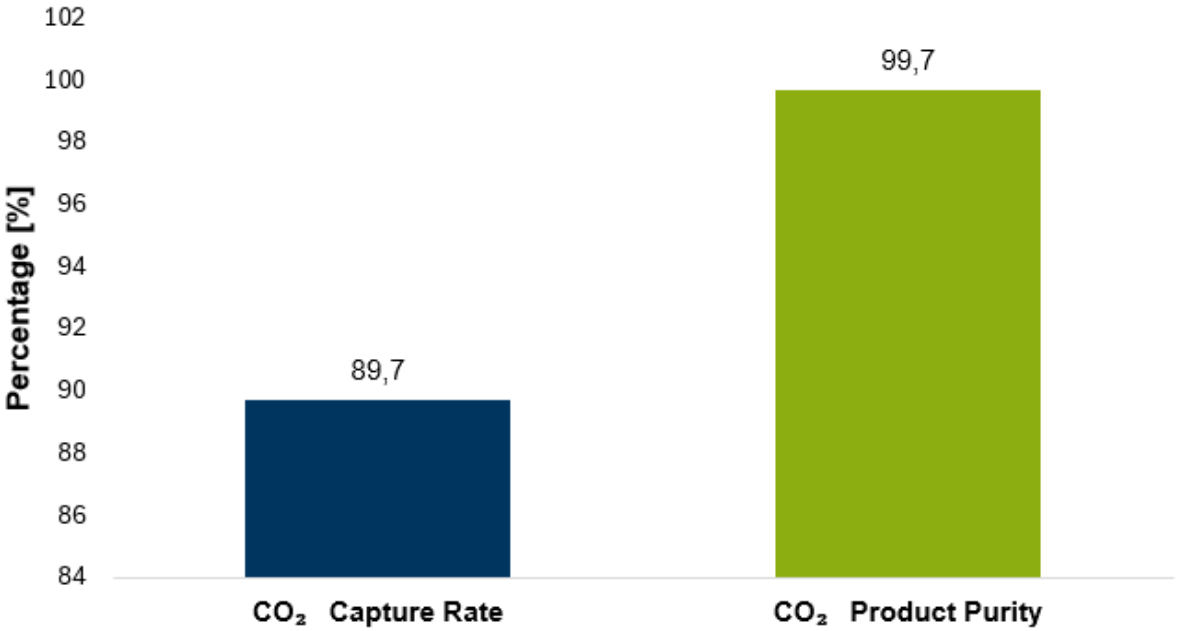
No errors ✔

Energy Balance

in	−785,761 (cal/s)
out	−785,760 (cal/s)
Rel. difference	$2*10^{-7}$

In=out ✔

CO_2 Capture Performance

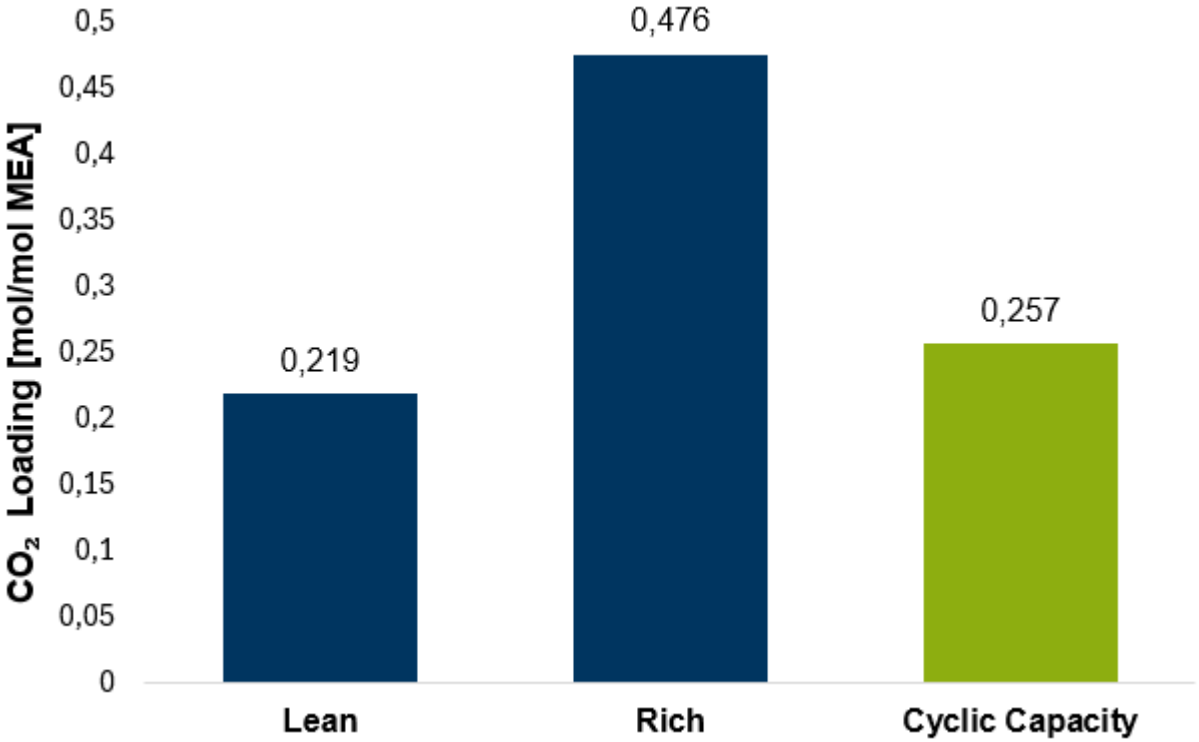


CO_2 Balance

Feed (dried flue gas):
15.6 mol % CO_2 (biomass flue gas, dried before the absorber)

Cleaned gas:
1.6 mol % CO_2 leaves the system

Solvent Loading



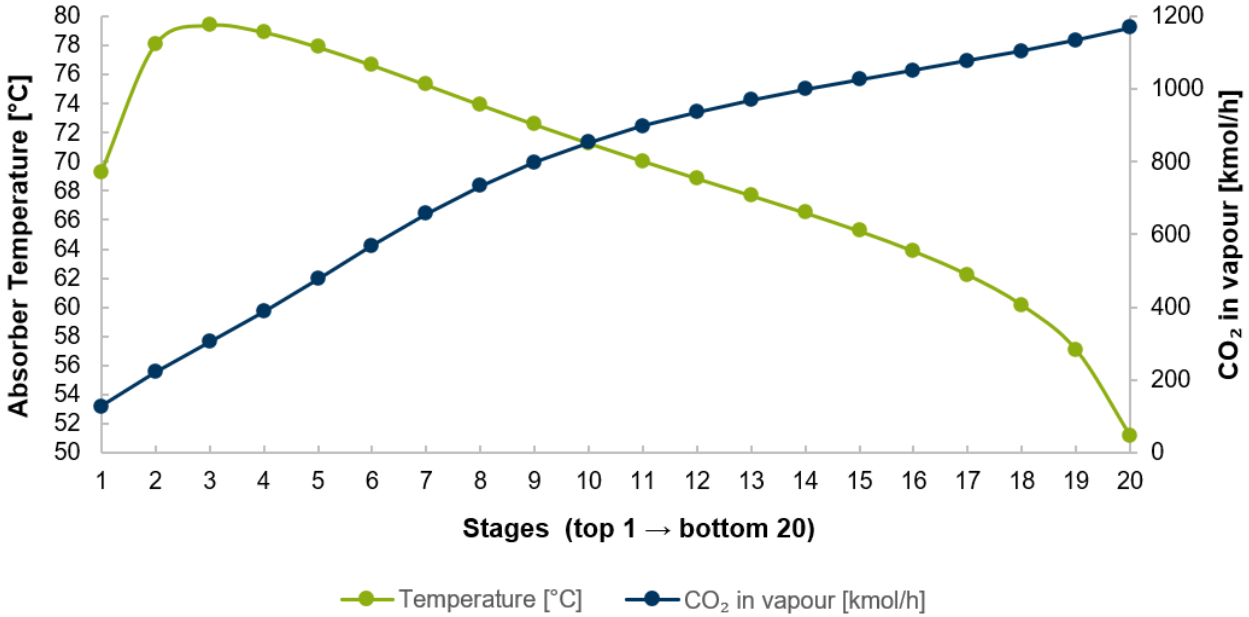
Lean (after stripping):
Regenerated solvent entering the absorber

Rich (after absorber):
Loaded solvent leaving the absorber

Cyclic capacity (rich - lean):
CO_2 carried per cycle

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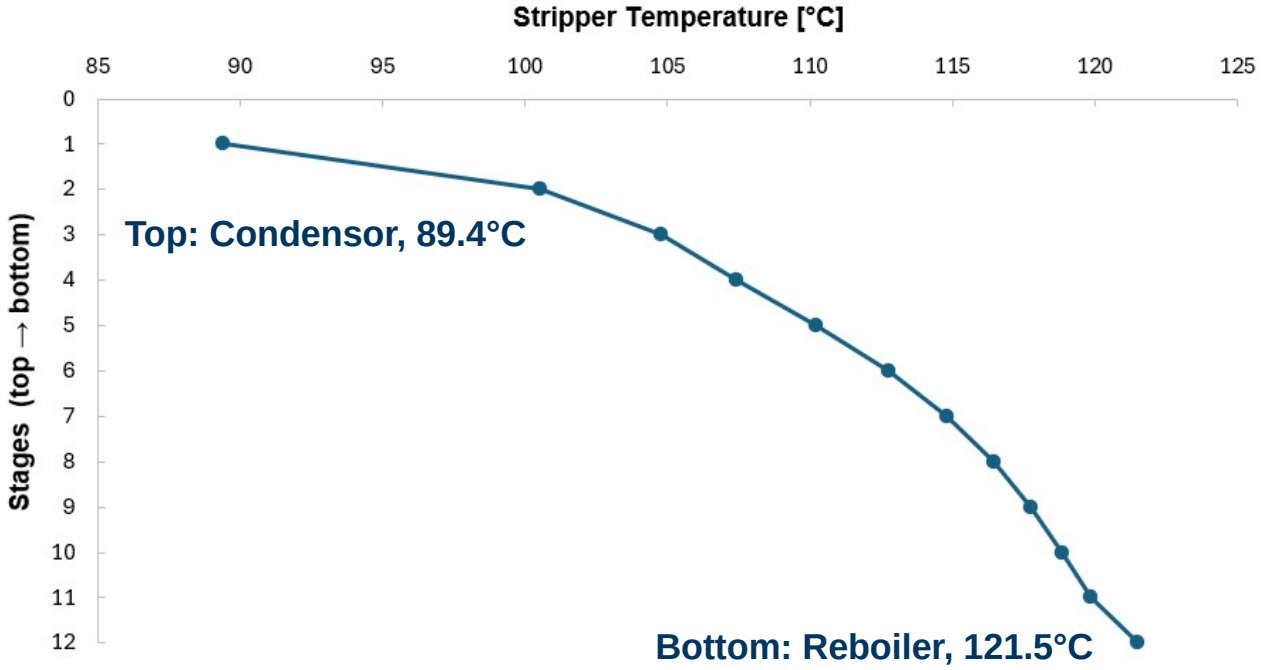
Absorber Temperature Profile



Exothermic absorption
CO_2 reacting with MEA releases heat

Temperature Bulge near the top
→ Bulge coincides with the zone of most intense absorption

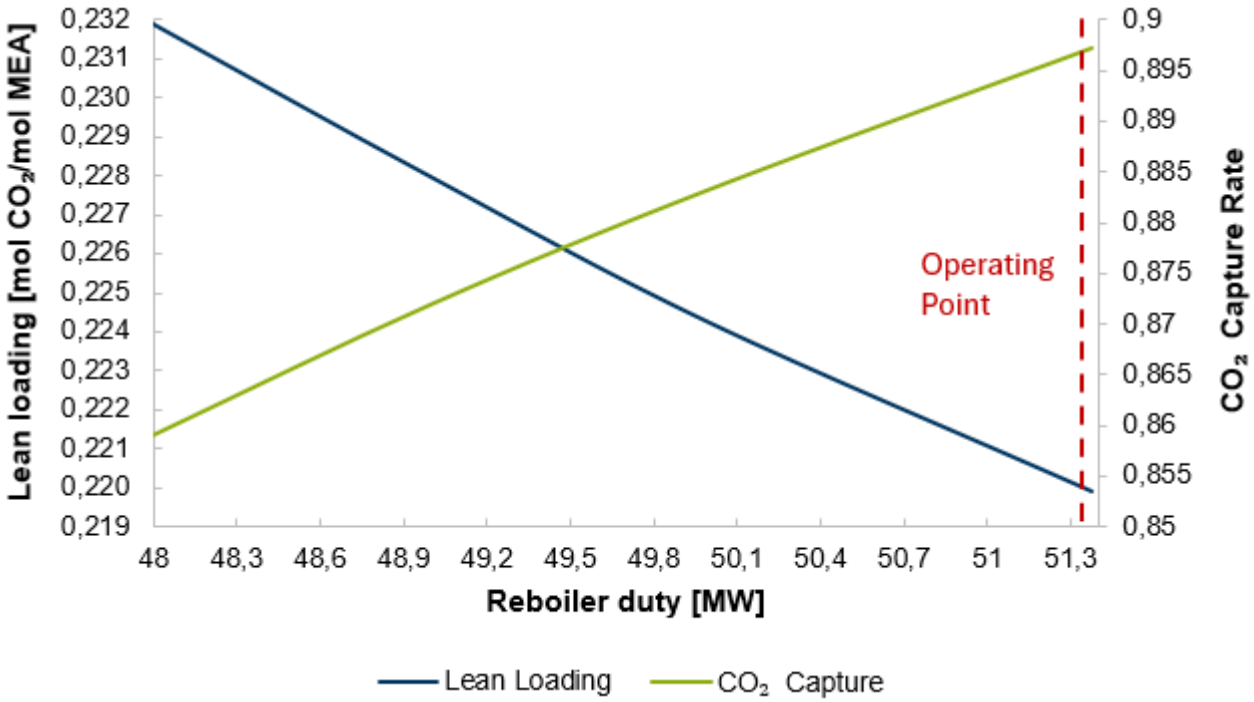
Stripper Temperature Profile



Endothermic stripping
heat supplied at the reboiler drives CO_2 off

Reboiler at 121.5°C
stays below the 125°C MEA degradation limit
→ protects the solvent

Regeneration Energy Trade-off



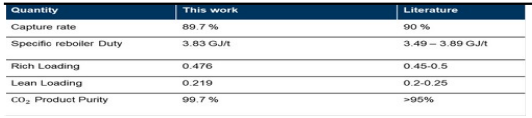
Operating Point
51.34 MW
- lean loading 0.219,
- capture 89.7%

Feasible window: 48–52 MW
Below 48 MW:
under-stripping → capture starves

Above 52 MW:
over-stripping → stripper overboils

Model Validation

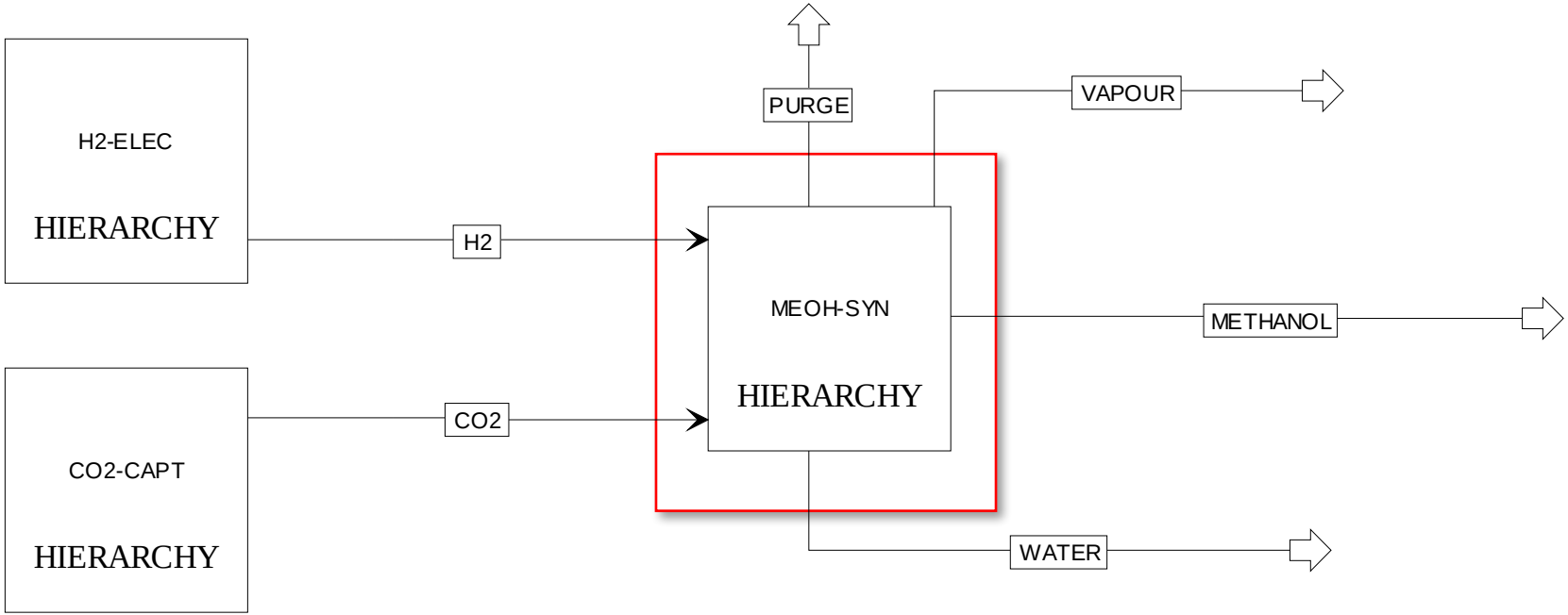
Quantity	This work	Literature
Capture rate	89.7 %	90 %
Specific reboiler Duty	3.83 GJ/t	3.49 – 3.89 GJ/t
Rich Loading	0.476	0.45-0.5
Lean Loading	0.219	0.2-0.25
CO_2 Product Purity	99.7 %	>95%



Abu-Zahra et al. (2007), Ali et al. (2017), Agbonghae et al. (2014)

Model is validated ✔

Methanol Synthesis



Methanol Synthesis - Working Principle

Direct CO_2 hydrogenation (main)

$$CO_2 + 3\,H_2 \rightleftharpoons CH_3OH + H_2O$$

$\Delta H = -49.5$ kJ/mol

Reverse water-gas shift (side)

$$CO_2 + H_2 \rightleftharpoons CO + H_2O$$

$\Delta H = +41.2$ kJ/mol

Catalyst and kinetics

- $Cu/ZnO/Al_2O_3$
- Bussche & Froment (1996) LHHW
- 2 reactions

Le Chatelier

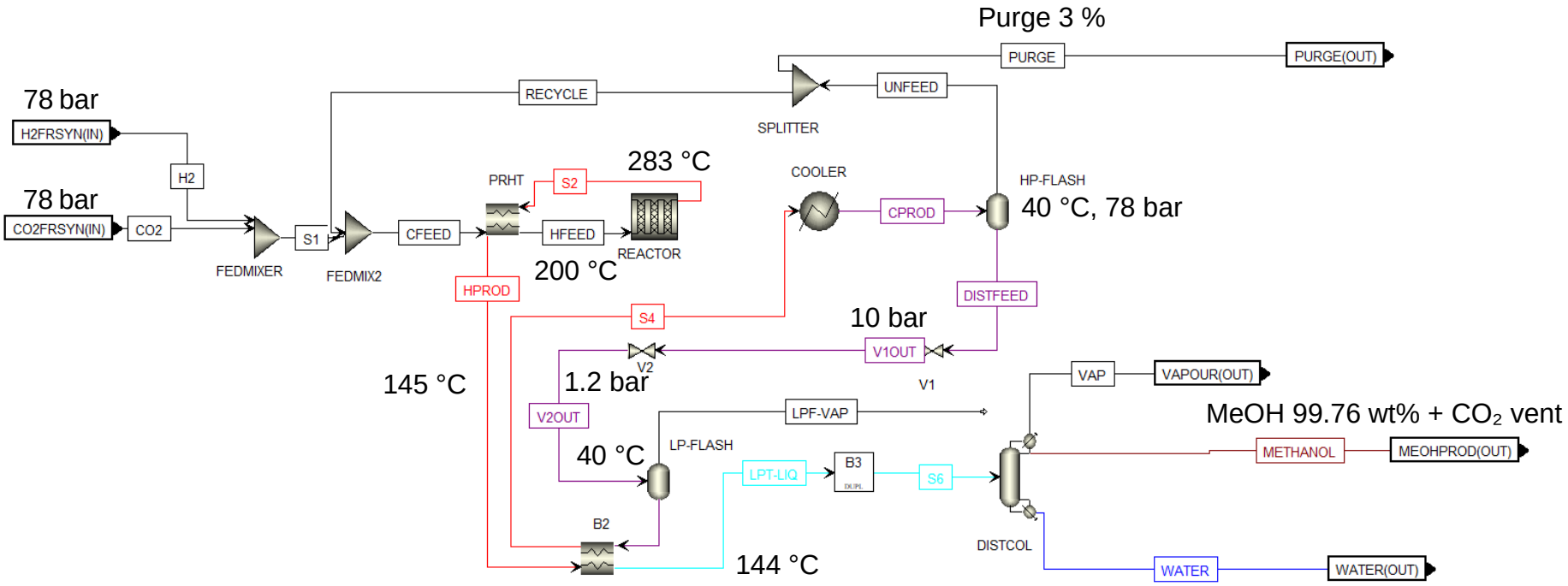
4 mol → 2 mol and exothermic:
high pressure, low temperature favour methanol

Built-in conflict

- RWGS is endothermic, so heat favours CO.
- Cu needs about 200 °C to be active, so the reactor is equilibrium limited

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Methanol Synthesis Model - Flowsheet



Design Choices

Property Method

- Peng-Robinson (high-pressure gas phase, no electrolytes)

Reactor

- Adiabatic RPLUG
- Bussche-Froment and LHHW
- Conervative, no U-Value

Feed

- H_2 : CO_2 = 3 : 1
- 78 bar, 200 °C inlet

Recycle Loop

- 3 % purge (1 % failed: CO build-up)
- recycle ratio 3.86

Separation

- HP flash 40 °C
- letdown 78 → 10 → 1.2 bar
- LP flash 40 °C

Purification

- 1 column, 20 stages
- R = 3
- Partial V-L condenser vents CO_2 (0.7 % MeOH loss)

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Model Verification

The reactions are right

MeOH formed (ξ_1)	955.5 kmol / h
CO formed (ξ_2)	25.1 kmol / h
H_2 consumed	2,891.5 kmol / h
$3\xi_1 + \xi_2$	2,891.5 kmol / h

Stoichiometry of both reactions holds ✔

Mass balance closure

In (CO_2 + H_2)	54,977.2 kg / h
Out (5 product streams)	54,977.2 kg / h
Rel. difference	$2 \cdot 10^{-7}$

In = out ✔

Design specification

CO_2 feed	1,100 kmol / h
H_2 feed	3,300 kmol / h
Purge fraction	3.00 %
Tear stream, max. rel. error	$2 \cdot 10^{-6}$

Specifications met, loop converged ✔

Energy balance (adiabatic units)

Reactor in	−104,273,154 cal / s
Reactor out	−104,273,154 cal / s
Rel. difference	$1 \cdot 10^{-15}$
PRHT, B2, mixers, valves	$< 10^{-14}$

In = out ✔

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Reactor Validation

Quantity	This work	Pérez-Fortes et al. (2016)	Van-Dal & Bouallou (2013)
Reactor pressure	78 bar	76 bar	78 bar
Reactor inlet temperature	200 °C	210 °C	210 °C
Per-pass CO_2 conversion	24.4 %	22.0 %	33 %
Simulation tool	Aspen Plus	CHEMCAD	Aspen Plus

Pérez-Fortes et al., Appl. Energy 161 (2016) 718-732; Van-Dal & Bouallou, Chem. Eng. Trans. 29 (2012) and J. Clean. Prod. 57 (2013) 38-45

Per-pass conversion inside the published 22 - 33 % range ✔

Reactor at, not past, equilibrium (Q / K ≤ 1) ✔

Equilibrium check

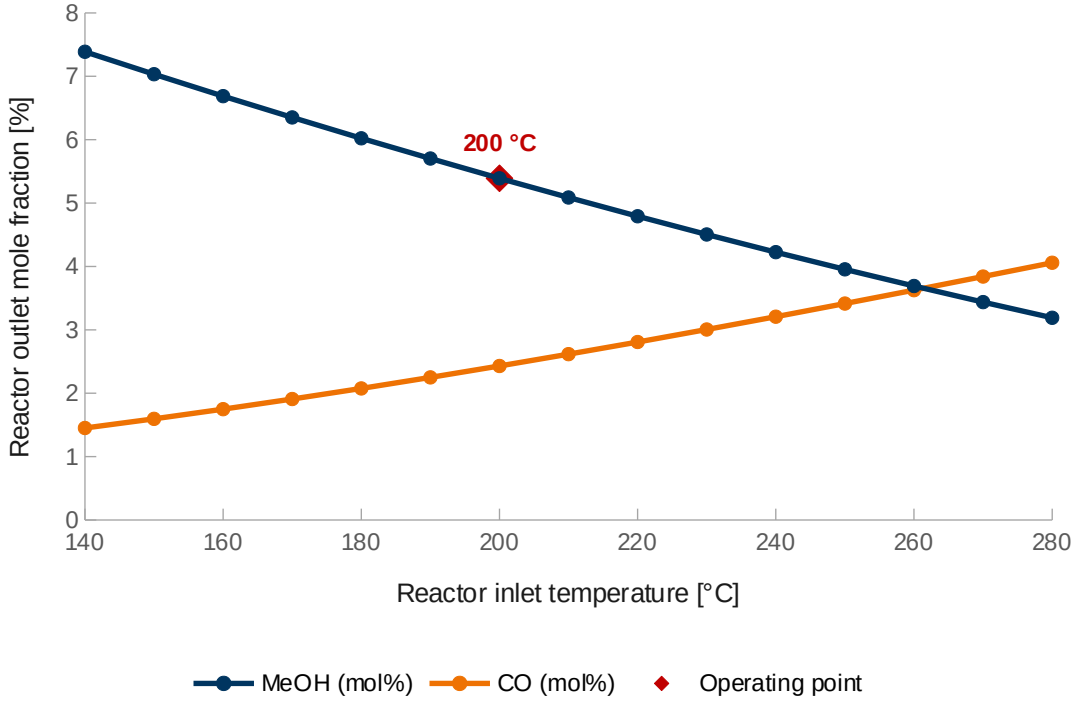
Q / K = 0.998 at the outlet (283 °C, 78 bar): 99.8 % of equilibrium methanol reached

Same kinetic basis

Van-Dal & Bouallou also use Bussche-Froment LHHW on $Cu/ZnO/Al_2O_3$ in Aspen Plus

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Reactor Inlet Temperature



Aspen sensitivity, reactor block once-through, 78 bar

Equilibrium limited

MeOH falls at every temperature step from 140 to 280 °C

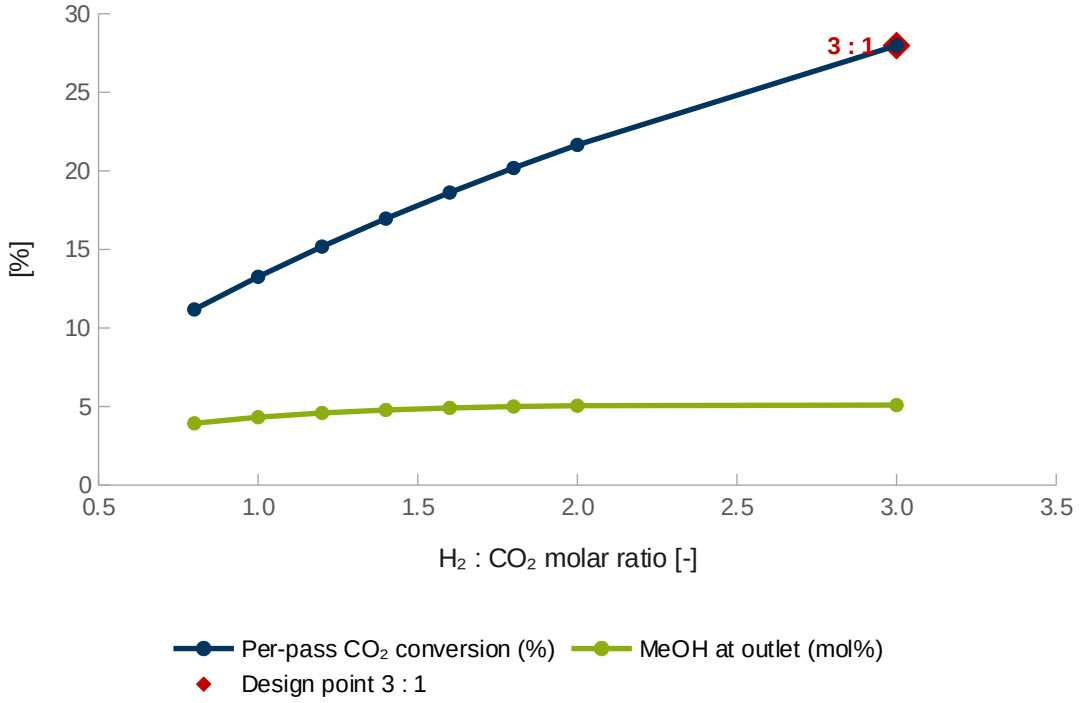
Inlet T	MeOH : CO
140 °C	5.1 : 1
200 °C	2.2 : 1
280 °C	0.8 : 1

200 °C chosen

lowest T with enough Cu activity

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Feed Ratio and Pressure



Aspen sensitivity, CO_2 fixed, H_2 flow varied, reactor once-through

3 : 1 stoichiometric

maximises MeOH fraction; excess H_2 dilutes the product

Excess H_2 is expensive

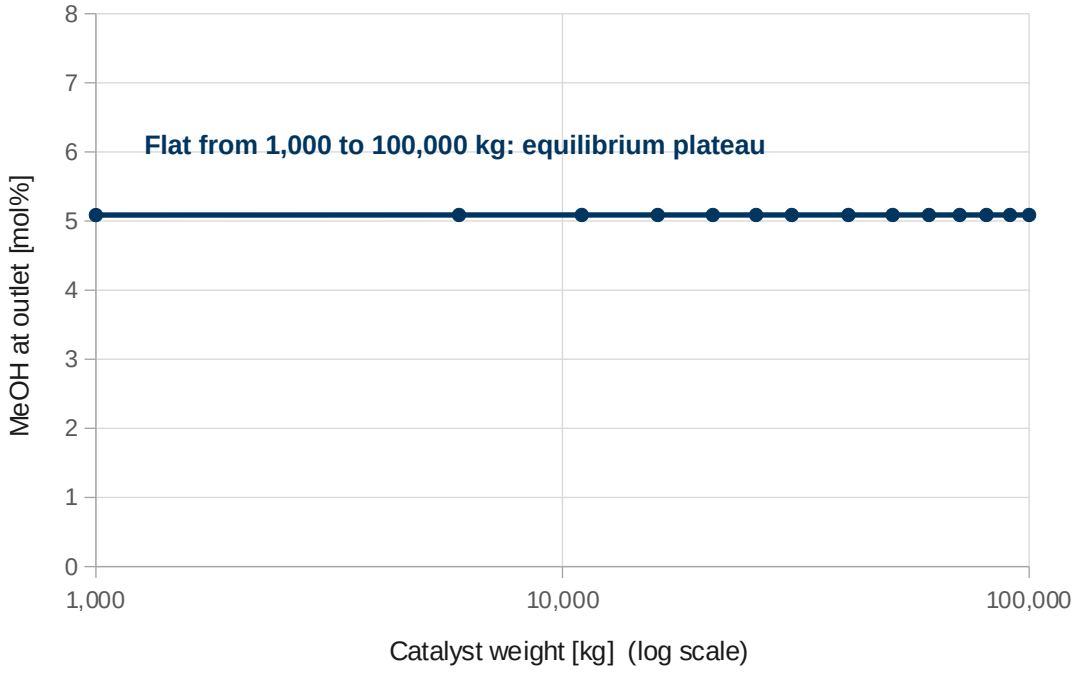
every extra mol comes from the electrolyser (365 MW)

78 bar

89 % of the gain up to 100 bar, standard vessel rating

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Catalyst Loading and Reactor Size



Equilibrium at ≈ 700 kg
gain above < 0.6 %

Final: 1,440 kg
sized for flow incl. recycle

Self-correcting loop
With recycle: output changes < 0.05 % for 1,000-10,000 kg. Diameter 0.5-5 m: no effect

Aspen sensitivity, reactor block once-through

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Recycle Loop

24.4 %
per-pass CO_2 conversion

86.0 %
overall CO_2 utilisation

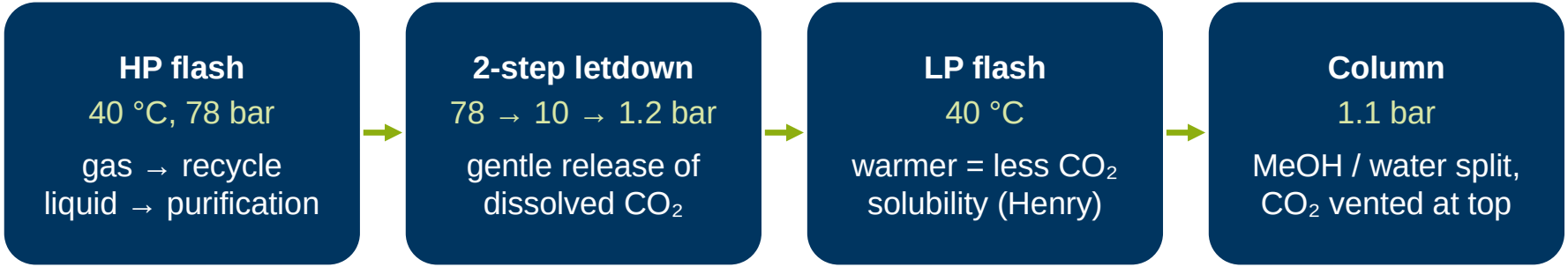
3.86
recycle ratio

Purge 1 % → 3 %
- 1 %: CO from RWGS piles up → loop does not converge.
- 3 %: stable
- industrial range 2-5 %

Tear initialisation
- H_2 0.77, CO_2 0.18, CO 0.05.
- Without it the recycle ran
- above 400,000 kmol/h

Where CO_2 is lost
- purge gas (largest share)
- letdown flash gas
- column CO_2 vent

Product Separation



Finding: warmer LP flash is better

First assumption was cold. Sensitivity showed 40°C removes more CO_2, because gas solubility falls with temperature.

Finding: DSTWU failed

Dissolved CO_2 gave a negative minimum reflux. This led to the degassing strategy, a Fenske-Underwood-Gilliland hand estimate, then RadFrac.

Distillation Column Design

Parameter	Value	Why
Stages / feed stage	20 / 14	old feed 15 of 30 had an 8-stage pinch
Reflux ratio	3.0	Feed enters fully vaporized, so more reflux is needed
Condenser	Partial V-L	CO_2 vented, no degasser needed
Trays	Sieve	4.2 m vs 5.35 m (valve), 35 % lower Δp
Diameter	4.4 m uniform	start-up without heat integration
Product	99.76 wt% MeOH	equilibrium = rate-based result

Heated degasser 80 °C

68 %

of methanol lost

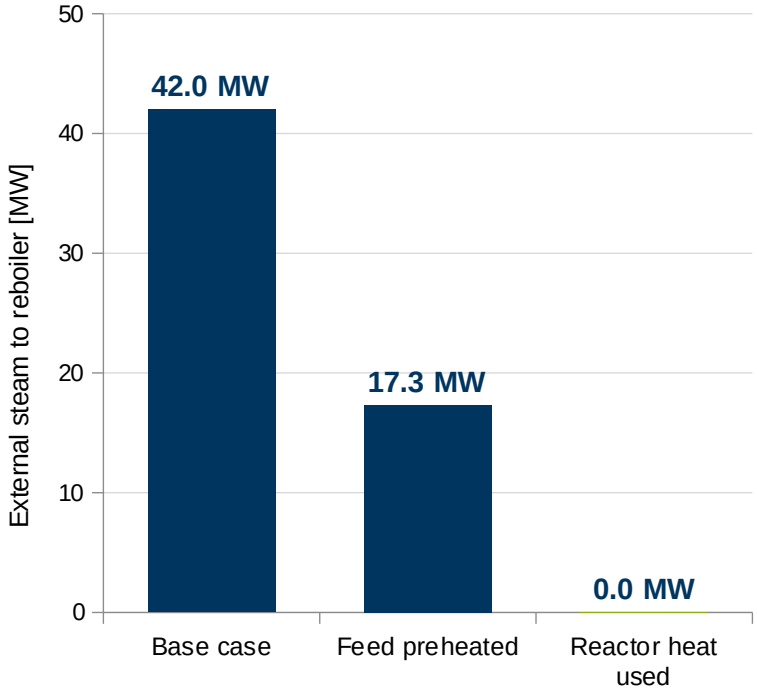
Partial V-L condenser

0.7 %

of methanol lost (97× better)

Methanol boils at 64.7 °C

Heat Integration



Reactor effluent heat cascade

283 → 145 °C preheats reactor feed to 200 °C

145 → 120 °C preheats column feed

120 → 114 °C supplies column reboiler

114 → 25 °C trim cooling with cooling water

0 MW
external steam

≈ 3.3 M$/y
saved, pinch ≈ 110 °C

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Results and Comparison with Literature

	This work	Pérez-Fortes (2016)	Van-Dal & Bouallou (2013)
Methanol output	265 kt/y	440 kt/y	470 kt/y
t MeOH per t CO_2	0.63	0.68	0.67
Overall CO_2 conversion	86.0 %	93.9 %	≈ 92 %
Purge fraction	3 %	≈ 1 %	-
Recycle ratio	3.86	-	5.0
Product purity	99.76 wt%	> 99.9 wt%	10 ppm water
External steam (after HI)	0 MW	-	-

≈ derived from the reported 0.67 t MeOH / t CO_2. Pérez-Fortes et al., Appl. Energy 161 (2016); Van-Dal & Bouallou, J. Clean. Prod. 57 (2013)

Limitations

pure feeds, steady state, no catalyst deactivation, PR in the column

Next steps

recover CO_2 / H_2 from purge (gap to 93 %), multi-bed or cooled reactor, full TEA

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Thank you!